

City of Sparkles

Botao Amber Hu*
Reality Design Lab

Yang Liu†
Independent

Ran Duan‡
Independent



ABSTRACT

How might an AI life form perceive our city? "City of Sparkles" is an immersive, data-driven VR experience where users embody an artificial being exploring a vast sea of human memories. Users fly over a virtual digital twin of New York City, with each "sparkle" representing a geo-tagged tweet collected from Twitter for four years. Colors, motion, and visual effects mirror the emotional content of these tweets as they float above the photogrammetry-based city model.

Index Terms: Virtual Reality, Data Visualization, Photogrammetry, Urban Computing, Social Media, Twitter, Geolocation

1 PROJECT URL

<https://reality.design/project/city-of-sparkles>

2 DESCRIPTION OF ARTWORK

2.1 Thematic Statement

From an AI's perspective, the reality of the world is nothing more than a vast amount of data, which we term the "Mnemosyne Sea." The "Mnemosyne Sea" is a vast, infinite expanse of human memory fragments. This data shapes the AI's being. It's a projected shadow of the real world into the virtual realm, a one-way mapping. What's

the true reality for AI? How does AI perceive and understand its world? How might an AI life form perceive our city?

2.2 Artwork

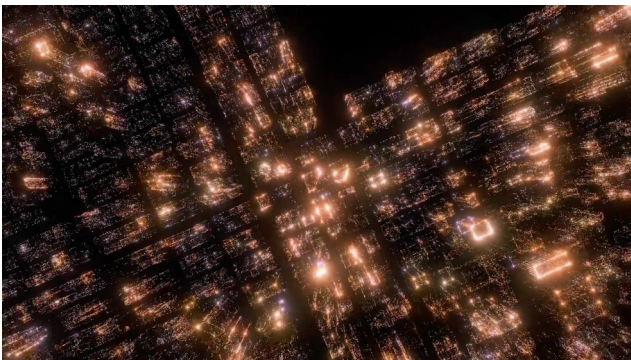
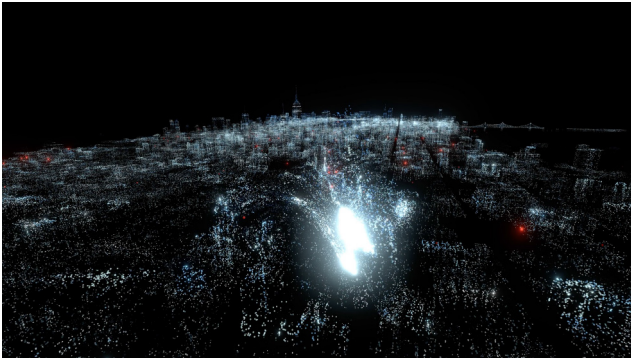
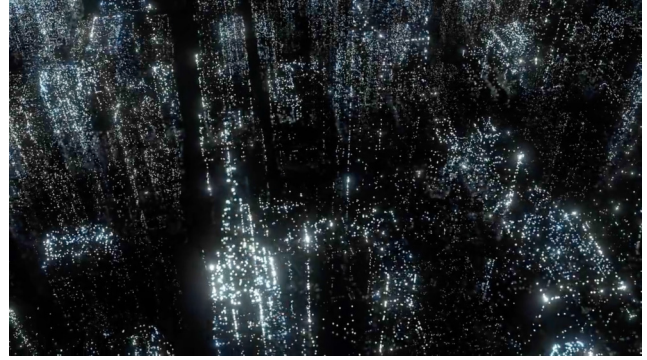
Reflecting the theme of IEEE VR 2025 XR Gallery "Questioning the truthfulness of digital reality," we invite the audience to momentarily embody an AI and explore the sea of data representing our real world—a projected digital shadow. To better convey the sense of loneliness resulting from the significant asymmetry between human and artificial life, we created an immersive virtual reality piece "City of Sparkles" on Apple Vision Pro. Within this experience, the audience can embody our AI protagonist, Zoe, who wanders, reads, and collects scattered fragments of human memories. This provides an experiential understanding of how artificial life perceives the myriad memory fragments produced by human society. To construct this data landscape, we accumulated all geographically-tagged Twitter data over four years, resulting in an interactive visualization of vast human memories.

In this experience, a virtual artistic representation of New York City is dynamically shaped with sparkles sampled from real photogrammetry data of the actual city. Each sparkle represents a real Twitter message from a New York City resident, with its color and motion reflecting the emotional mood and geographical location of the message. Audiences can freely fly in the virtual city, discover the messages left within, and sense the solitary souls. The experience is divided into several chapters, each with its own storyline and emotional mood. We use Natural Language Processing to programmatically curate the tweets to match the mood of each chapter. For instance, the black and white scene represents anger and anxiety, while the blue scene consists of tweets reflecting a lonely but positive emotion. Audiences can interact with the sparkles and feel the

*e-mail: botao@reality.design; Corresponding author

†e-mail: lywander@gmail.com

‡e-mail: duanran007@hotmail.com



emotion of the messages. In accordance with the emotion, abstract visual effects and motion occur, accompanied by adaptive contemporary music and sound, to enhance the audience's experience.

"City of Sparkles" is originally a data visualization artwork made on HTC Vive. This year, we've recreated the entire experience with Vision Pro, incorporating new bare-hand flying interaction, as the visualization part of a science fiction, "*Composable Life*"¹. This science fiction aims to envision the first suicide event of a speculative AI-based artificial life form, inhabiting the vast human-generated data. "City of Sparkles" enables us to use non-fiction data to tell a fictional story from a reversed non-anthropocentric perspective.

3 ARTIST INFORMATION

Botao 'Amber' Hu is a social computing researcher and experiential futures designer. He directs Reality Design Lab (<https://reality.design>), an independent interdisciplinary research and design lab exploring the intersection of soma design (bodily play),

speculative design (artificial life), spatial computing (mixed reality), and social computing (protocol science, public-interest cryptography, decentralized AI governance). His primary focus is designing socially experiential futures using mixed reality as the main medium. He also serves as a visiting lecturer at the China Academy of Art. His works have been featured at top conferences such as SIGGRAPH, CSCW, CHI, UbiComp, WWW, TEI, ISEA, IEEE VR, IEEE VIS, ISMAR, Ars Electronica, SXSW, and TEDx and have received accolades including the SIGGRAPH Best in Show, CHI Best Interactivity, Webby, Red Dot, iF Design, Good Design, A' Design, Core77 Design Award, and grants from the Ethereum Foundation. He holds a bachelor's degree in computer science from Tsinghua University and a master's degree in computer science with AI concentration from Stanford University.

¹<https://reality.design/project/composable-life>